

Hugo JARKOFF

Machine Learning Software Engineer (MSc.)

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ML Software Engineer with 5+ years of experience building and deploying ML systems at scale.

EXPERIENCE

- Addactis (formerly NamR, acquired in 2025)** Paris, France
ML/CV Software Engineer 2024 - present
Responsible for production ML systems powering aerial imagery processing and SaaS products:
 - Refactored the team’s ML training framework, reducing time from data acquisition to production release by over 50%.
 - Redesigned the aerial imagery preprocessing pipeline with serverless Cloud Run APIs, cutting legacy codebase size down and reducing data integration time for ML workflows.
 - Refactored the aerial imagery inference pipeline into a distributed GPU batch-processing system with PostgreSQL-coordinated workers, achieving an 80% faster runtime and 90% lower compute cost on large-scale datasets.
 - Designed, built, and deployed an AI agent platform (FastAPI + Pydantic-AI) integrated into production SaaS insurance products.
- Finegrain** Paris, France
ML/CV Software Engineer 2024 (4 months)
State-of-the-art generative model fine-tuning for image editing:
 - Fine-tuned foundation models (SAM, Stable Diffusion) using adapters (LoRA, IP-Adapter) for image editing tasks.
 - Conducted continuous literature review in generative modeling.
 - Contributed to the team’s high-performance OSS micro-framework for foundation model adaptation (Refiners).
- Invoxia** Paris, France
R&D Machine Learning Engineer 2021 – 2024
Led multiple end-to-end ML projects from research to deployment:
 - Designed a CV-based signal processing model to estimate canine respiratory rate from accelerometer data; achieved 98.5% accuracy (preprint available).
 - Researched and adapted neural architectures (CNNs, Vision Transformers), focusing on model compression for embedded deployment.
 - Deployed ML models to 10k+ production devices via Kubernetes and AWS SageMaker, with real-time monitoring (Grafana, Prometheus).
- Sopra Steria (for NavBlue - Airbus)** Toulouse, France
End-of-Studies Team Project 2019 – 2020
 - Developed an airport image segmentation and classification system using fully convolutional neural networks (UNet).

EDUCATION

- ISAE-SUPAERO** Toulouse, France
Engineering Degree (MSc.) 2016 – 2020
 - *Major in Machine Learning*: Deep Learning (Computer Vision, NLP), RL, MLOps.
 - *Minor in Advanced Mathematics*: Bayesian statistics, HPC, optimization.
- Lycée Louis-le-Grand** Paris, France
Preparatory Classes 2014 – 2016
 - *PCSI - PC**: Mathematics - Physics - Chemistry - Computer Science.

TECHNICAL SKILLS

- **Programming**: advanced knowledge of Python (multithreaded programming, async programming); intermediate knowledge of C++ / C / Bash.
- **Deep Learning**: PyTorch, TensorFlow, Keras.
- **MLOps & Infrastructure**: Docker, Kubernetes, Terraform, NVIDIA Triton, PyTriton, CI/CD (GitLab), experiment tracking (WandB, ClearML).
- **Databases**: PostgreSQL, PostGIS.
- **Cloud Platforms**: Google Cloud Platform (GCP), Amazon Web Services (AWS).

LANGUAGES

- **French**: Native language.
- **English**: Full professional proficiency.
TOEFL ITP: 633/677.

PERSONAL PROJECTS AND DIVERSE INTERESTS

- **Language Models Training**: RapGPT: Trained a Transformer-based language model from scratch (architecture, training loop, and deployment) to generate French rap lyrics; deployed live demos on Hugging Face (with open weights).
- **Web Development**: Developed a simple, lightweight (700KB running), multiprocess (fork) HTTP server in C++.
- **Rock Climbing**: Experienced climber (20+ years of practice). Designed and built a custom finger-force measurement device (ESP32 soldered with deformation gauge, embedded with C software).